Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Microfluidic lung-on-chip platform for pre-screening of inhalation toxicity focusing on pulmonary barrier penetration and direct tissue damage			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ) , Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio・Medical			
	Medium Category	Small Category	Medium Category	Small Category
	Bio-processes / Bio-Equipment	Bio-Analytical Instruments		
Concept of Proposed Technology	This proposed technology aims to develop a lung-on-chip platform based on air–liquid interface (ALI) culture to pre-screen inhalation toxicity of candidate substances prior to GLP inhalation toxicity studies. The platform focuses on identifying substances that penetrate the pulmonary barrier or cause direct damage to lung epithelial tissue by using microfluidic lung chips integrated with quantitative readouts such as barrier integrity, permeability, and tissue damage indicators. The main purpose of the system is not to replace GLP inhalation toxicity studies, but to serve as a cost-effective and time-saving preliminary screening method to identify high-risk substances in advance and thus reduce animal testing.			

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Rationale for Joint R&D Project	GLP inhalation toxicity studies are characterized by high cost, long duration, and limited testing capacity, resulting in significant bottlenecks in safety evaluation pipelines. A joint R&D approach is required to integrate advanced lung in vitro models and microfluidic technologies. Also validate pre-screening results through comparison with GLP inhalation toxicity data and ensure regulatory relevance and applicability of the developed platform. The Collaboration between a Korean SME specializing in organ-on-chip commercialization, Fraunhofer IBMT with strong expertise in lung in vitro models and microfluidis, and a GLP-certified testing organization (Korea Conformity Laboratories KCL) is essential to achieve practical and regulatory-aligned outcomes.
Key Words	1. Lung-on-Chip 2. Inhalation Toxicity 3. Air–Liquid Interface 4. Microfluidics 5. Pre-screening Platform

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (7)
R&D Objectives	- Develop an ALI-based lung-on-chip platform for inhalation toxicity pre-screening - Establish quantitative criteria for pulmonary barrier penetration and direct tissue damage - Validate correlation between lung-on-chip results and GLP inhalation toxicity data - Demonstrate applicability of the platform as a pre-screening tool in safety assessment workflows			
R&D Target Specifications	- Stable ALI lung culture under microfluidic flow for ≥ 14 days - Reproducibility of barrier integrity measurements (CV ≤ 20%) - Quantitative permeability assessment using tracer compounds - Detection of direct lung tissue damage with sensitivity ≥ 80% - Correlation coefficient ≥ 0.6 between chip-based results and GLP inhalation toxicity outcomes			
R&D Plan	Year 1 - Design and fabrication of ALI-based lung-on-chip prototype - Establishment of standardized lung epithelial culture conditions - Selection of key endpoints for barrier integrity and tissue damage Year 2 - Optimization of microfluidic exposure and measurement protocols - In vitro inhalation toxicity testing using reference substances - Preliminary correlation analysis with available GLP inhalation data Year 3 - Expanded validation using selected test substances - Comparative analysis with GLP inhalation toxicity studies - Definition of applicability domain and pre-screening decision criteria - Preparation for commercialization and service integration			

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	- Development and optimization of lung in vitro models under ALI conditions - Design of microfluidic lung-on-chip architecture and exposure scenarios - Execution of in vitro inhalation toxicity testing and data analysis - Contribution to interpretation of biological relevance and model limitations - Close coordination with KCL on the interpretation of data relating to GLP studies
Expected R&R of Korean SME Partner	- Commissioning of GLP studies and close cooperation with KCL - Product-oriented design and manufacturing of lung-on-chip devices - Standardization and reproducibility control of chip production - System integration and prototype scale-up - Coordination of joint R&D activities and commercialization planning

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	- Establishment of a practical lung-on-chip platform for inhalation toxicity pre-screening - Reduction of animal testing and unnecessary GLP inhalation toxicity studies through early risk filtering - Improved efficiency and capacity utilization of inhalation toxicity testing facilities - Creation of new service and product opportunities for Korean SMEs in the global safety assessment market - Strengthening international collaboration in advanced in vitro toxicology technologies

6. References

(1) OECD, Guidance Document on Good In Vitro Method Practices (GIVIMP)

(2) OECD, Test Guideline Programme – Advanced In Vitro Models

(3) FDA, Predictive Toxicology Roadmap

(4) OECD Test Guideline No. 413: "Subchronic Inhalation Toxicity: 90-day Study"

(5) Kohl, Y., Biehl, M., Spring, S., Hesler, M., Ogourtsov, V., Todorovic, M., Owen, J., Elje, E., Kopecka, K., Moriones, O. H., Bastús, N.G., Simon, P., Dubaj, T., Rundén-Pran, E., Puntes, V., William, N., von Briesen, H., Wagner, S., Kapur, N., Mariussen, E., Nelson, A., Gabelova, A., Dusinska, M., Velten, T., Knoll, T.: “Microfluidic In Vitro Platform for (Nano)Safety and (Nano) Drug Efficiency Screening”. Small 2021, 2006012. DOI: 10.1002/smll.202006012

(6) Kohl, Y., Hesler, M., Drexel, R., Kovar, L., Dähnhardt-Pfeiffer, S., Selzer, D., Wagner, S., Lehr, T., von Briesen, H., Meier, F.: “Influence of Physicochemical Characteristics and Stability of Gold and Silver Nanoparticles on Biological Effects and Translocation across an Intestinal Barrier-A Case Study from In Vitro to In Silico”. Nanomaterials 2021, May 21,11(6), 1358. DOI: 10.3390/nano11061358

(7) Müller, M., Kohl, Y., Germann, A., Wagner, S., Zimmermann, H., von Briesen, H.; “Alveolar epithelial-like cell differentiation in a dynamic bioreactor: a promising 3D-approach for the high-throughput generation of lung cell types from human induced pluripotent stem cells”. In vitro models 2023; DOI: 10.1007/s44164-023-00052-1

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization